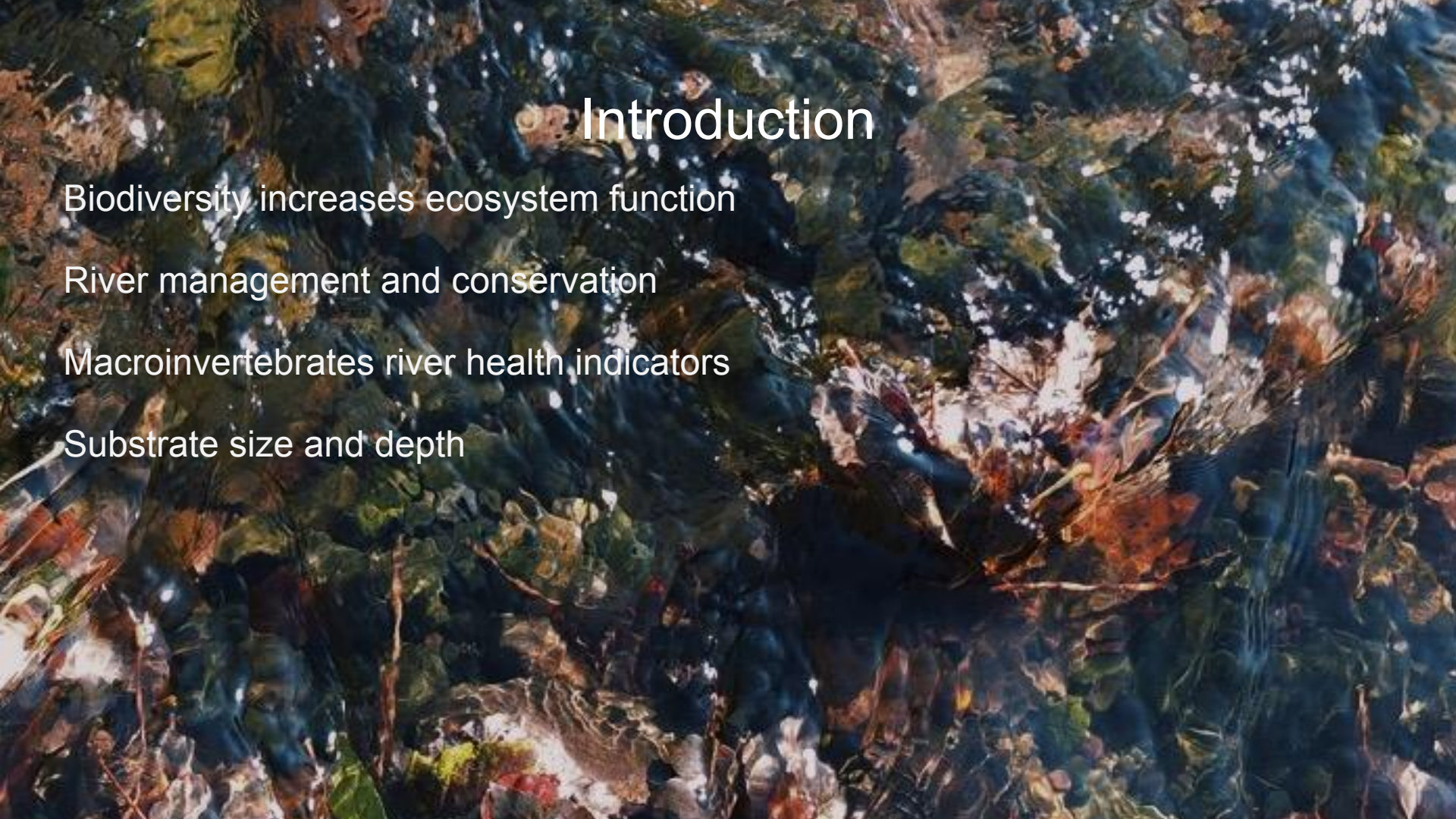


Freshwater Invertebrate Richness and Spatial Structure

A photograph of a forest stream with a large fallen log in the foreground. The water is flowing over the log, creating a small pool and ripples. The surrounding forest has trees with green and yellow autumn foliage. The text is overlaid on the upper half of the image.

How do substrate size and river depth influence the richness of freshwater macroinvertebrates in the Braid Burn stream?

Eira, Faye, Susanna, Holli, Katja, Hayward

The background of the slide is a photograph of a riverbed. The water is clear, revealing a rocky bottom covered with green algae and moss. Numerous brown, fallen leaves are scattered across the rocks and in the water, creating a textured and naturalistic scene. The lighting is bright, suggesting daylight.

Introduction

Biodiversity increases ecosystem function

River management and conservation

Macroinvertebrates river health indicators

Substrate size and depth

Scientific question and hypotheses

How do substrate size and stream depth influence the richness of freshwater macroinvertebrates in the Braid Burn stream?

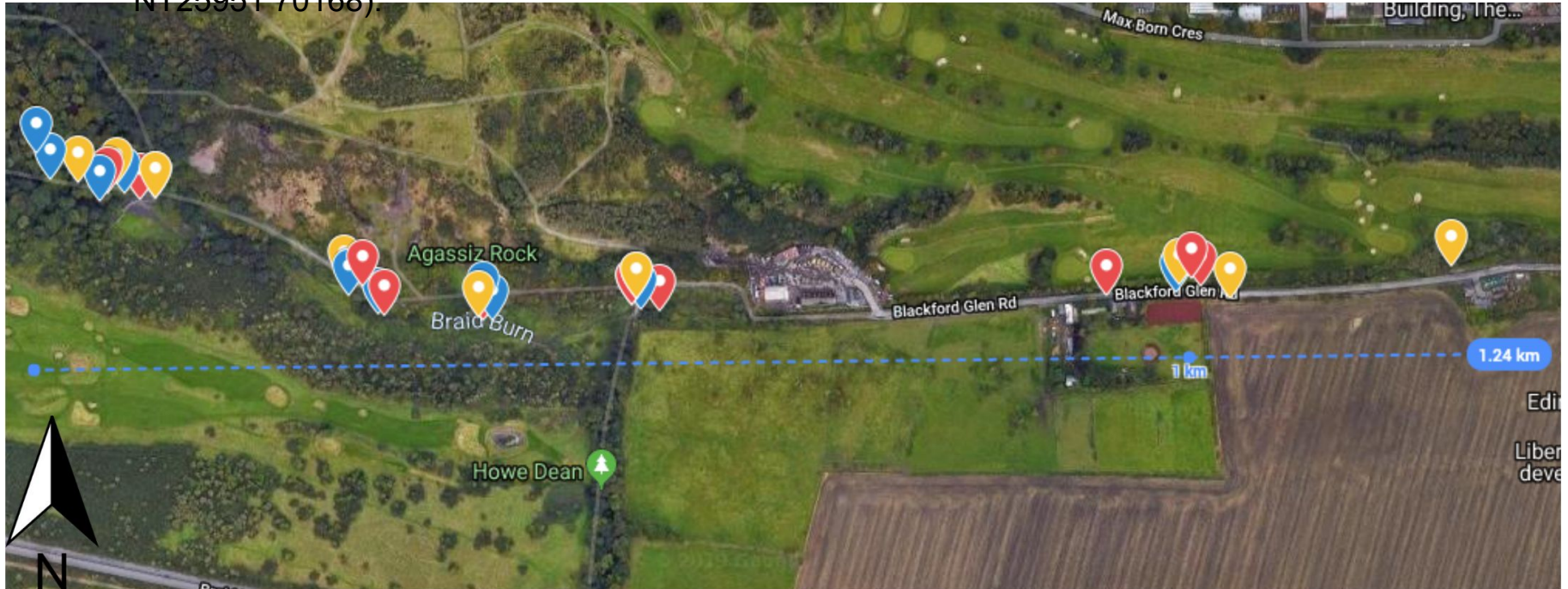
Expectations:

- Richness increases with substrate size
- Richness increases with depth



Study site

- Braid Burn stream (semi-managed) near the Hermitage of Braid, in Edinburgh (grid reference: NT25951 70168).



● SMALL ● MEDIUM ● LARGE

Habitat details

- **Along asphalt road (east):** mixed deciduous hedgerows, herbaceous stream banks (grasses). Crop field (South), managed golf course (North). 0-10% Canopy cover: <10%.
- **Hermitage of Braid public park (west):** mixed deciduous woodland- herbaceous banks (grasses, beech, birch and hazel). Canopy cover: 20-40%. Increasing woodland density with progression westward.
- **Topography:** relief progressed from flat/sloping fields to steeper valley slopes within the woodland.
- **Conditions:** consistently fair/overcast, little wind, temperature range between 11-13 °C. Light precipitation during data collection on 16th.





Method

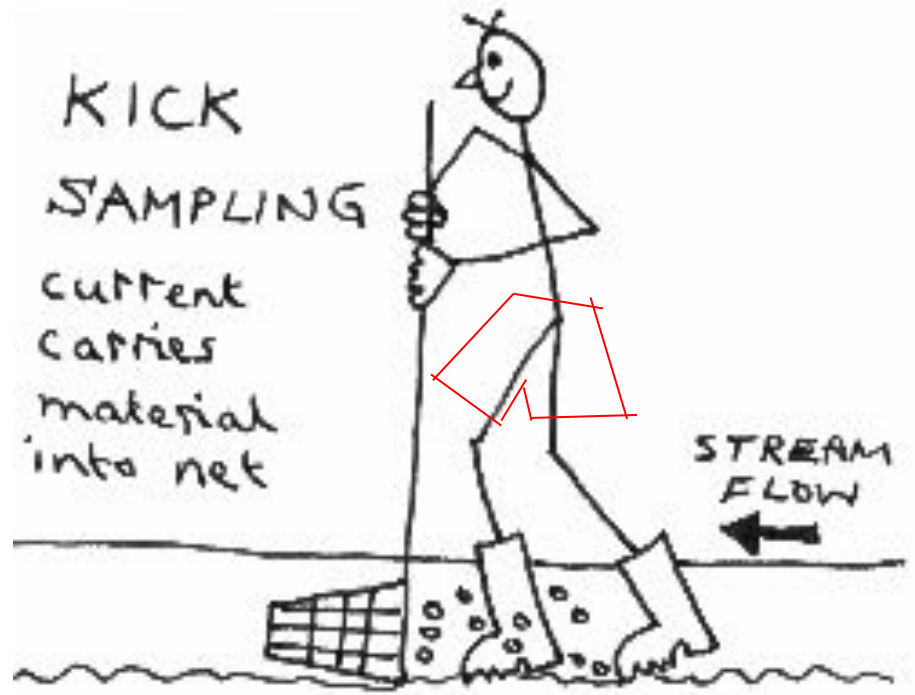
- 16.10.19-30.10.19 10-13
- The Braid Burn
- Accessible
- Same sub sizes, 20 m away from each other
- Allocation to specific tasks - consistency



Substrate sizes



Kick-sampling method



River depth



Identifying invertebrate species



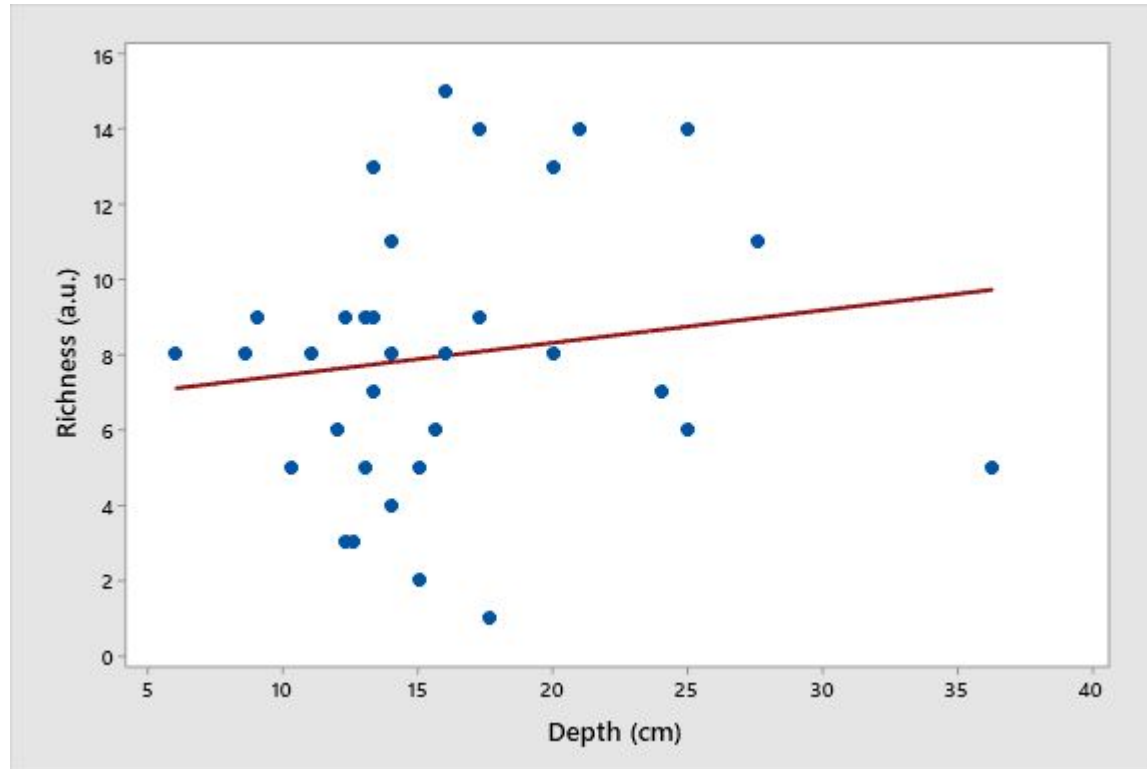
Data analysis

- Minitab
- Normality tests
- Regression analysis: depth and richness
- One-way ANOVA: richness and substrate size



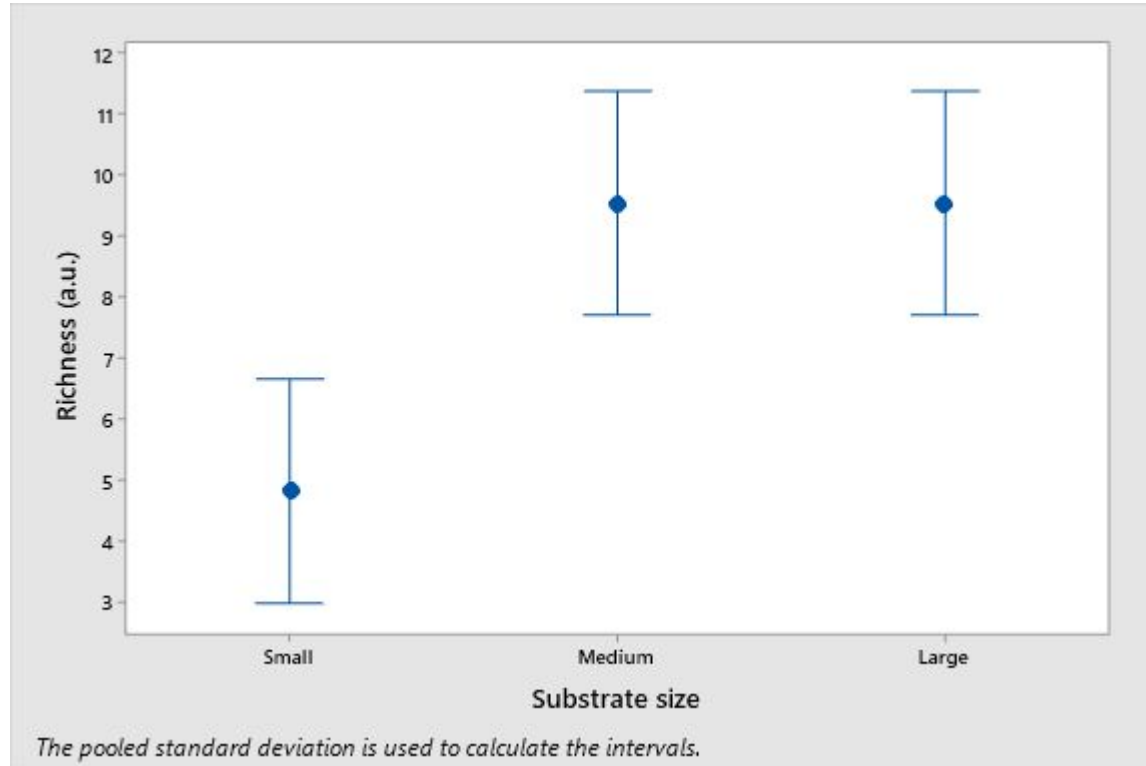
Results and discussion

Regression analysis (regression, $F=0.66$, $P=0.423$)



Results and discussion

Richness and substrate size (One-way ANOVA $F=9.20$, $P=0.001$)



Improvements and Future Studies

- Improvements to Design
- Practical improvements
- Other possible factors to take into account
- Future studies

Conclusion

- How does substrate size and stream depth influence the richness of freshwater macroinvertebrates in Braid Burn stream?
- We accepted the null hypothesis 'there will no significant relationship between depth and species richness' i.e. relationship was not significant
- Significant difference between richness for small and medium, and small and large substrate sizes.

Vertebrate discovery - Splishy and Splashy



References

Duan, X., Wang, Z. and Tian, S. (2008). Effect of streambed substrate on macroinvertebrate biodiversity. *Frontiers of Environmental Science & Engineering in China*, 2(1), pp.122-128.

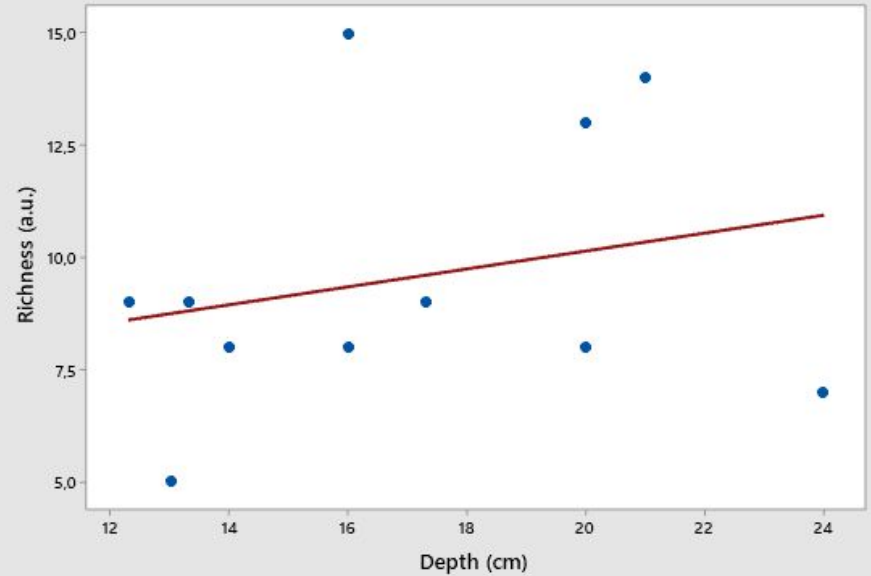
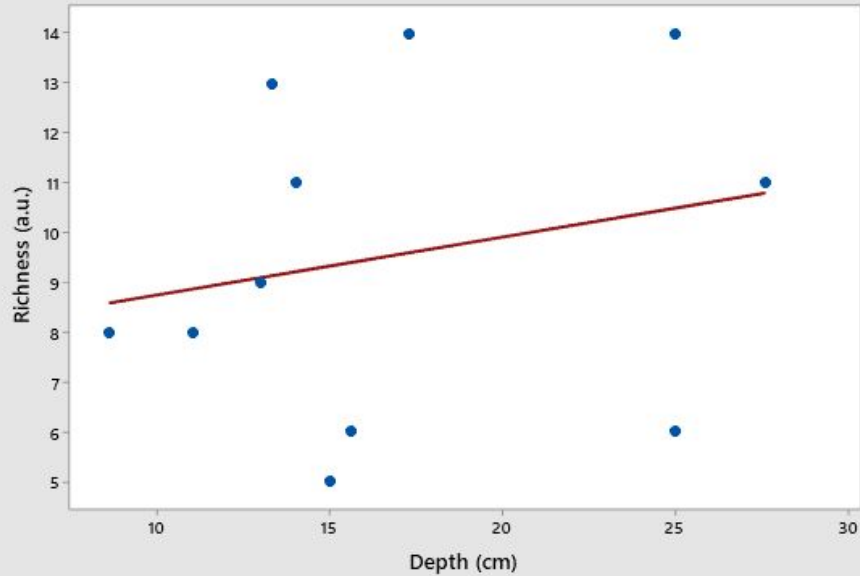
Webb, D. (2010). Freshwater shrimp (*Palaemonetes australis*) as a potential bioindicator of crustacean health. *Environmental Monitoring and Assessment*, 178(1-4), pp.537-544.

Williams, D. D. and J. H. Mundie (1978). Substrate size selection by stream invertebrates and the influence of sand. *Limnology and Oceanography* 23(5) 1030-1033.

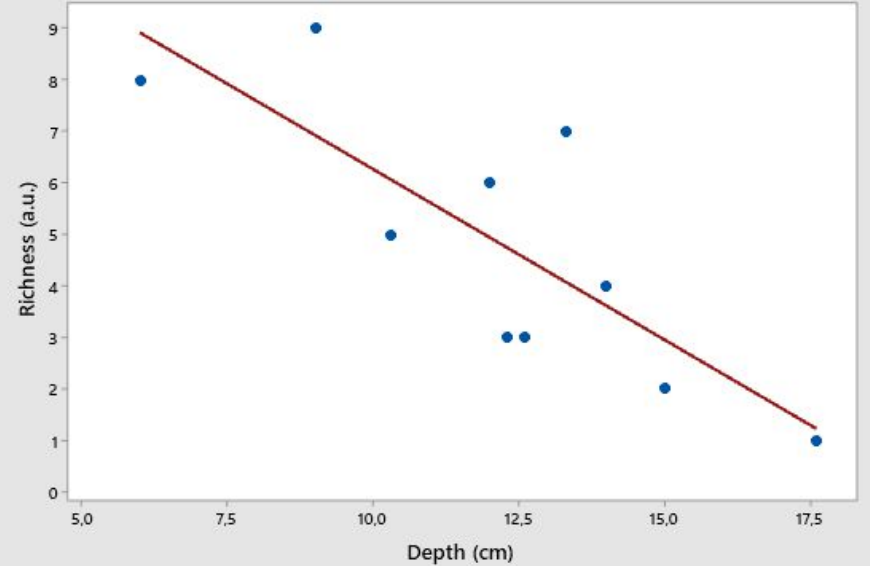
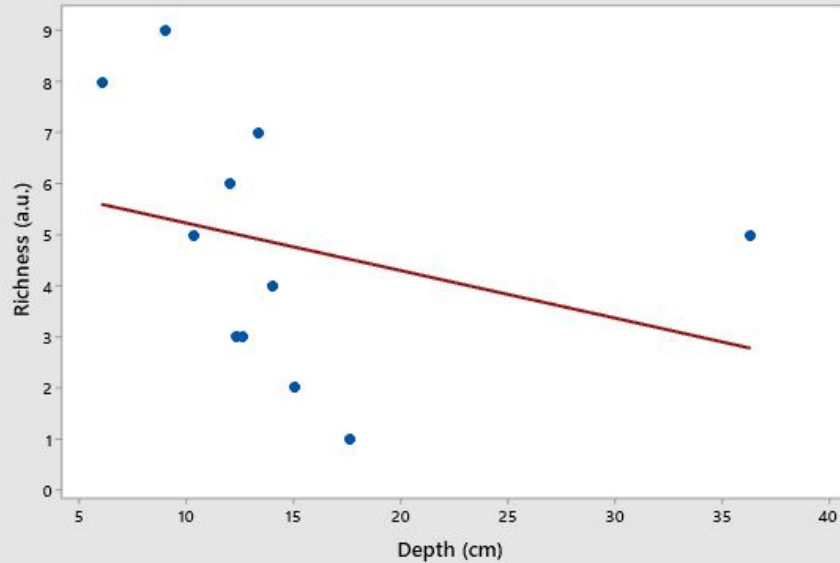
A group of people are crossing a shallow stream in a wooded area. The sun is high in the sky, creating a bright lens flare. The water is shallow and rocky, with people using poles to cross. The surrounding vegetation is dense and green. The word "Questions?" is overlaid in large white text across the center of the image.

Questions?

Medium and large: regression analysis



Small substrate: regression analysis



Normality tests

